In [72]:
```
import pandas as pd
import numpy as np
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import IsolationForest
from sklearn.metrics import silhouette_score
from sklearn.neighbors import LocalOutlierFactor
from scipy import stats
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

# Set plot style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")
plt.rcParams['figure.dpi'] = 300
plt.rcParams['savefig.dpi'] = 300

print("Medicaid OB/GYN Fee Schedule Analysis (1989-2012)")
print("=" * 80)

# Load dataset and assess data quality
df = pd.read_excel("obgyn-medicaid-fee-schedules.xlsx")

print("DATA QUALITY CHECK")
print(f"Total obs: {len(df)}")
print(f"States: {df['State'].nunique()}")
print(f"NaN by CPT: {df[['CPT_59400', 'CPT_59510', 'CPT_59610']].isnull().su

# Check for time dimension
time_cols = [col for col in df.columns if any(x in col.lower() for x in ['ye
print(f"Time columns detected: {time_cols}")

# STEP 1: Filter for COMPLETE observations (ALL 3 CPT columns required)
df_clean = df.dropna(subset=['CPT_59400', 'CPT_59510', 'CPT_59610']).copy()
print(f"Clean obs (all 3 CPTs): {len(df_clean)} ({len(df_clean)/len(df)*100:

# STEP 2: MARKUP CALCULATION + Clinical bounds filter
df_clean['markup'] = df_clean['CPT_59510'] / df_clean['CPT_59400']
df_clean = df_clean[df_clean['markup'].between(0.5, 3.0)].copy()
print(f"After markup filter: {len(df_clean)}")

# TIME DIMENSION ANALYSIS
print("\n" + "="*50)
print("ENHANCEMENT 1: TIME TREND ANALYSIS")
print("="*50)

if 'Year' in df_clean.columns:
    df_clean['Year'] = pd.to_numeric(df_clean['Year'], errors='coerce')
    years = df_clean['Year'].dropna().unique()
    print(f"Years available: {sorted(years)}")

    yearly_markup = df_clean.groupby('Year')['markup'].agg(['mean', 'std', '
```

```
    print("\nYearly Markup Trends:")
    print(yearly_markup)

    yearly_csection = df_clean.groupby('Year')['CPT_59510'].agg(['mean', 'st
    print("\nAnnual C-section price trends:")
    print(yearly_csection)
else:
    print("No explicit time column found. Analysis assumes cross-sectional c

# State-level markup analysis
print("\n" + "="*50)
print("State C-Section Markup Rankings (Top 15)")
print("="*50)
state_markup = df_clean.groupby('State')['markup'].agg(['mean', 'std', 'cour
state_markup.columns = ['Avg_Markup', 'Volatility', 'Observations']
state_markup['Markup_Pct'] = (state_markup['Avg_Markup'] * 100).round(0).ast
state_markup = state_markup.sort_values('Avg_Markup', ascending=False)
print(state_markup[['Avg_Markup', 'Markup_Pct', 'Volatility', 'Observations'

# STEP 3-6: K-MEANS CLUSTERING
features = df_clean[['CPT_59400', 'CPT_59510', 'CPT_59610']].copy()
print(f"\nFeatures shape: {features.shape}")
print(f"Any NaNs in features: {features.isnull().any().any()}")

scaler = StandardScaler()
features_scaled = scaler.fit_transform(features)

print("\nK-MEANS CLUSTERING (Scaled + Complete Data)")
max_k = min(8, len(features_scaled) // 3 + 1)
inertias = []
for k in range(1, max_k):
    kmeans_temp = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans_temp.fit(features_scaled)
    inertias.append(kmeans_temp.inertia_)

optimal_k = 3
kmeans = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
df_clean['cluster'] = kmeans.fit_predict(features_scaled)

cluster_means = df_clean.groupby('cluster')[['CPT_59400', 'CPT_59510', 'CPT_
print(f"Optimal clusters: {optimal_k}")
print("\nCluster Centers (Unscaled):")
print(cluster_means.round(0))


# CLUSTER INTERPRETATION WITH DESCRIPTIVE LABELS

print("\n" + "="*50)
print("CLUSTER INTERPRETATION & LABELING")
print("="*50)

# Data-driven cluster labeling based on C-section price (CPT_59510)
csection_means = cluster_means['CPT_59510'].sort_values()
cluster_order = csection_means.index.tolist()

print("Cluster ranking by C-section reimbursement (lowest to highest):")
```

```
for i, cluster_id in enumerate(cluster_order):
    print(f"  Cluster {cluster_id}: ${csection_means[cluster_id]:,.0f} - {cl

# Assign descriptive labels based on actual centroids
cluster_profiles = {
    cluster_order[0]: "Low-Cost States",      # Lowest C-section price
    cluster_order[1]: "Medium-Cost States",   # Medium C-section price
    cluster_order[2]: "High-Cost States"      # Highest C-section price
}

df_clean['cluster_label'] = df_clean['cluster'].map(cluster_profiles)
state_cluster_labels = df_clean.groupby(['cluster_label', 'State']).size().r

print(f"\nCluster label distribution:")
print(df_clean['cluster_label'].value_counts().sort_index())
print(f"\nSample state assignments:")
print(state_cluster_labels.head(10))

# VALIDATED OUTLIER DETECTION
print("\n" + "="*50)
print("ENHANCEMENT 2: VALIDATED OUTLIER DETECTION")
print("="*50)

iso_forest = IsolationForest(contamination=0.1, random_state=42)
df_clean['iso_outlier'] = iso_forest.fit_predict(features_scaled)

lof = LocalOutlierFactor(n_neighbors=20, contamination=0.1)
df_clean['lof_outlier'] = lof.fit_predict(features_scaled)

z_scores = np.abs(stats.zscore(features))
df_clean['z_outlier'] = (z_scores > 3).any(axis=1).astype(int) * 2 - 1

df_clean['consensus_outlier'] = ((df_clean['iso_outlier'] == -1) &
                                 (df_clean['lof_outlier'] == -1) |
                                 (df_clean['z_outlier'] == -1)).astype(int) *

n_consensus = sum(df_clean['consensus_outlier'] == -1)
print(f"Consensus outliers (2+ methods): {n_consensus} ({n_consensus/len(df_
print("Method agreement:")
print(pd.crosstab(df_clean['iso_outlier'], df_clean['lof_outlier'], margins=

consensus_outliers = df_clean[df_clean['consensus_outlier'] == -1]['State'].
print("\nTop consensus outlier states:", consensus_outliers.head(8).to_dict(

# COMPREHENSIVE VISUALIZATIONS (ACTIVE)
print("\n" + "="*50)
print("VISUALIZATIONS")
print("="*50)

# 1. STATE ANALYSIS (2x2)
fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle('Medicaid OB/GYN State Analysis', fontsize=16, fontweight='bold

top15 = state_markup.head(15).copy()
colors = plt.cm.Reds(np.linspace(0.3, 1, len(top15)))
axes[0,0].barh(range(len(top15)), top15['Avg_Markup'], color=colors, alpha=0
```

```
axes[0,0].set_yticks(range(len(top15)))
axes[0,0].set_yticklabels(top15.index, fontsize=10)
axes[0,0].set_xlabel('Markup Ratio')
axes[0,0].set_title('Top 15 States: Markup Ratio', fontweight='bold')
axes[0,0].grid(axis='x', alpha=0.3)

state_csection = df_clean.groupby('State')['CPT_59510'].mean().head(15)
axes[0,1].barh(range(len(state_csection)), state_csection.values, color=plt.
axes[0,1].set_yticks(range(len(state_csection)))
axes[0,1].set_yticklabels(state_csection.index, fontsize=10)
axes[0,1].set_xlabel('C-Section Price ($)')
axes[0,1].set_title('Top 15 States: C-Section Price', fontweight='bold')
axes[0,1].grid(axis='x', alpha=0.3)

axes[1,0].barh(range(len(top15)), top15['Volatility'], color=plt.cm.Oranges(
axes[1,0].set_yticks(range(len(top15)))
axes[1,0].set_yticklabels(top15.index, fontsize=10)
axes[1,0].set_xlabel('Volatility (Std Dev)')
axes[1,0].set_title('Top 15 States: Price Volatility', fontweight='bold')
axes[1,0].grid(axis='x', alpha=0.3)

axes[1,1].barh(range(len(top15)), top15['Observations'], color=plt.cm.Greens
axes[1,1].set_yticks(range(len(top15)))
axes[1,1].set_yticklabels(top15.index, fontsize=10)
axes[1,1].set_xlabel('Observations')
axes[1,1].set_title('Top 15 States: Sample Size', fontweight='bold')
axes[1,1].grid(axis='x', alpha=0.3)

plt.tight_layout()
plt.savefig('medicaid_state_analysis.png', bbox_inches='tight', dpi=300)
plt.show()

# 2. CLUSTER & OUTLIER ANALYSIS (2x2) - WITH LABELS
fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle('Cluster Analysis & Outlier Validation', fontsize=16, fontweigh

# Cluster centers with labels
(cluster_means[['CPT_59400', 'CPT_59510']].T
 .rename(index={'CPT_59400': 'Vaginal', 'CPT_59510': 'C-Section'})
 .plot(kind='bar', ax=axes[0,0], width=0.8))
axes[0,0].set_title('Cluster Centers by Procedure Type')
axes[0,0].set_ylabel('Average Price ($)')
axes[0,0].legend(title='Procedure', fontsize=10)
axes[0,0].grid(alpha=0.3)

# Elbow plot
axes[0,1].plot(range(1, max_k), inertias, 'bo-', linewidth=2, markersize=8)
axes[0,1].set_xlabel('Number of Clusters (k)')
axes[0,1].set_ylabel('Inertia')
axes[0,1].set_title('Elbow Method')
axes[0,1].grid(True, alpha=0.3)
axes[0,1].axvline(x=3, color='red', linestyle='--', label='Optimal k=3')
axes[0,1].legend()

# Outlier method comparison
outlier_methods = ['iso_outlier', 'lof_outlier', 'z_outlier', 'consensus_out
```

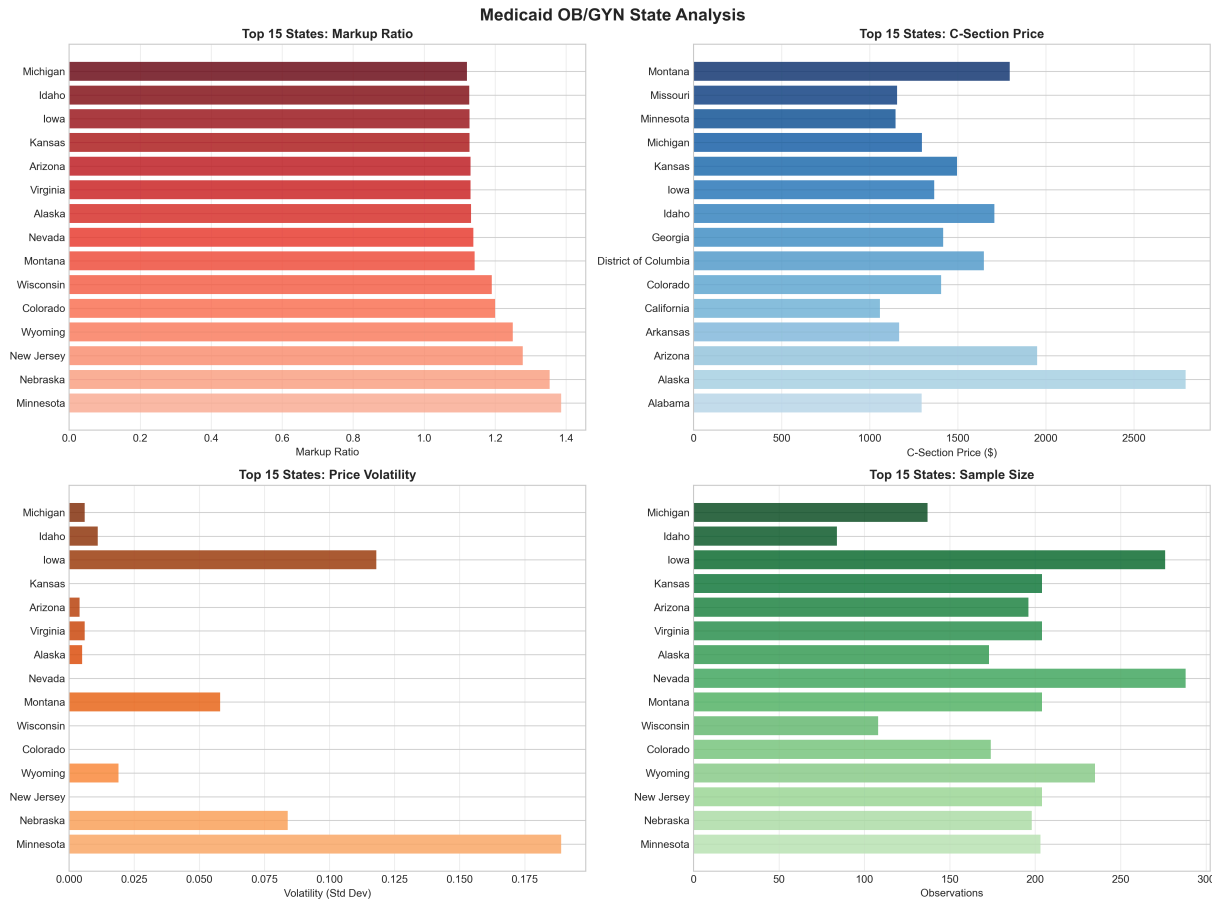
```
counts = [sum(df_clean[m] == -1) for m in outlier_methods]
axes[1,0].bar(outlier_methods, counts, color=['red', 'orange', 'blue', 'purp
axes[1,0].set_title('Outlier Detection Methods')
axes[1,0].set_ylabel('Outliers Detected')
axes[1,0].tick_params(axis='x', rotation=45)

# Price scatter by cluster WITH LABELS
colors = ['green', 'orange', 'red']  # Low, Medium, High cost
for i, color in enumerate(colors):
    subset = df_clean[df_clean['cluster'] == i]
    label = cluster_profiles.get(i, f"Cluster {i}")
    axes[1,1].scatter(subset['CPT_59400'], subset['CPT_59510'],
                     c=color, alpha=0.6, s=30, label=label)
axes[1,1].set_xlabel('Vaginal Delivery ($)')
axes[1,1].set_ylabel('C-Section ($)')
axes[1,1].set_title('Price Scatter by Cost Tier')
axes[1,1].legend()
axes[1,1].grid(alpha=0.3)

plt.tight_layout()
plt.savefig('medicaid_cluster_outliers.png', bbox_inches='tight', dpi=300)
plt.show()

# 3. STATISTICAL DISTRIBUTIONS
fig, axes = plt.subplots(2, 2, figsize=(15, 12))
fig.suptitle('Statistical Analysis', fontsize=16, fontweight='bold')

# Markup distribution
axes[0,0].hist(df_clean['markup'], bins=30, alpha=0.7, color='skyblue', dens
axes[0,0].axvline(df_clean['markup'].mean(), color='red', linestyle='--', li
axes[0,0].set_xlabel('Markup Ratio')
axes[0,0].set_ylabel('Density')
axes[0,0].set_title('Markup Distribution')
axes[0,0].legend()
axes[0,0].grid(alpha=0.3)

# Price distributions (overlaid)
axes[0,1].hist(df_clean['CPT_59400'], bins=25, alpha=0.7, color='green', lab
axes[0,1].hist(df_clean['CPT_59510'], bins=25, alpha=0.7, color='blue', labe
axes[0,1].hist(df_clean['CPT_59610'], bins=25, alpha=0.5, color='purple', la
axes[0,1].set_title('Price Distributions by Procedure')
axes[0,1].set_xlabel('Price ($)')
axes[0,1].legend()

# Correlation matrix
corr_matrix = df_clean[['CPT_59400', 'CPT_59510', 'CPT_59610', 'markup']].co
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', center=0, ax=axes[1,0]
axes[1,0].set_title('Correlation Matrix')

# Silhouette score
sil_scores = []
K_range = range(2, min(8, len(features_scaled)//3 + 1))
for k in K_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(features_scaled)
    sil_scores.append(silhouette_score(features_scaled, km.labels_))
```

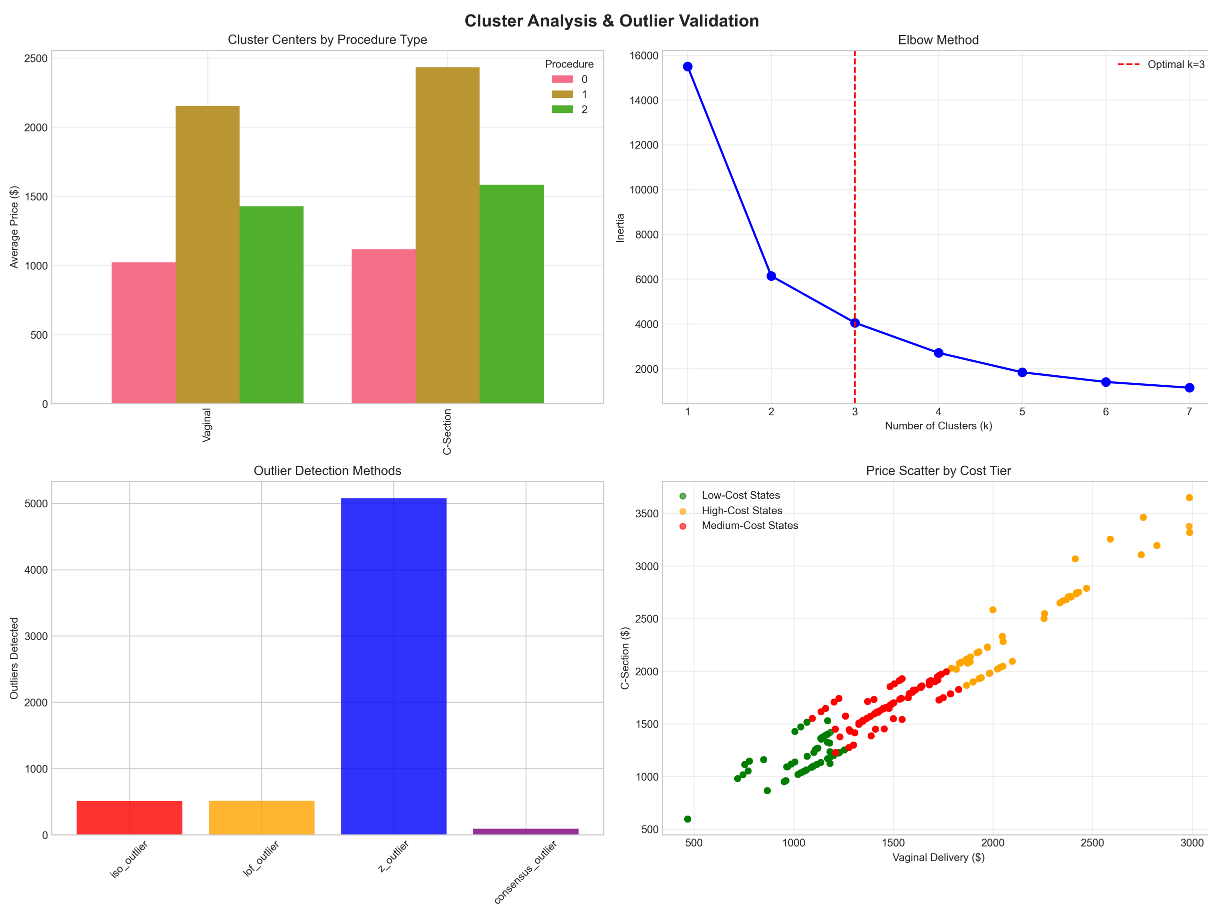
```
axes[1,1].plot(K_range, sil_scores, 'bo-', linewidth=2)
axes[1,1].set_xlabel('Number of Clusters')
axes[1,1].set_ylabel('Silhouette Score')
axes[1,1].set_title('Silhouette Analysis')
axes[1,1].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('medicaid_statistical.png', bbox_inches='tight', dpi=300)
plt.show()

# FINAL SUMMARY
print("\n" + "="*80)
print("ANALYSIS SUMMARY WITH CLUSTER INTERPRETATION")
print("="*80)
print(f"Valid observations: {len(df_clean)} across {df_clean['State'].nuniqu
print(f"Silhouette score: {silhouette_score(features_scaled, df_clean['clust
print(f"Consensus outliers: {n_consensus} ({n_consensus/len(df_clean)*100:.1
print(f"Top markup states: {list(state_markup.head(3).index)}")

print( "CLUSTER INTERPRETATION:")
for label, count in df_clean['cluster_label'].value_counts().sort_index().it
    avg_price = df_clean[df_clean['cluster_label'] == label]['CPT_59510'].me
    print(f"  {label}: {count:,} states (${avg_price:,.0f} avg C-section)")

print("VISUALIZATIONS GENERATED (300 DPI):")
print("   • medicaid_state_analysis.png")
print("   • medicaid_cluster_outliers.png")
print("   • medicaid_statistical.png")
print("\n   • df_clean now contains 'cluster_label' column for further analy
```

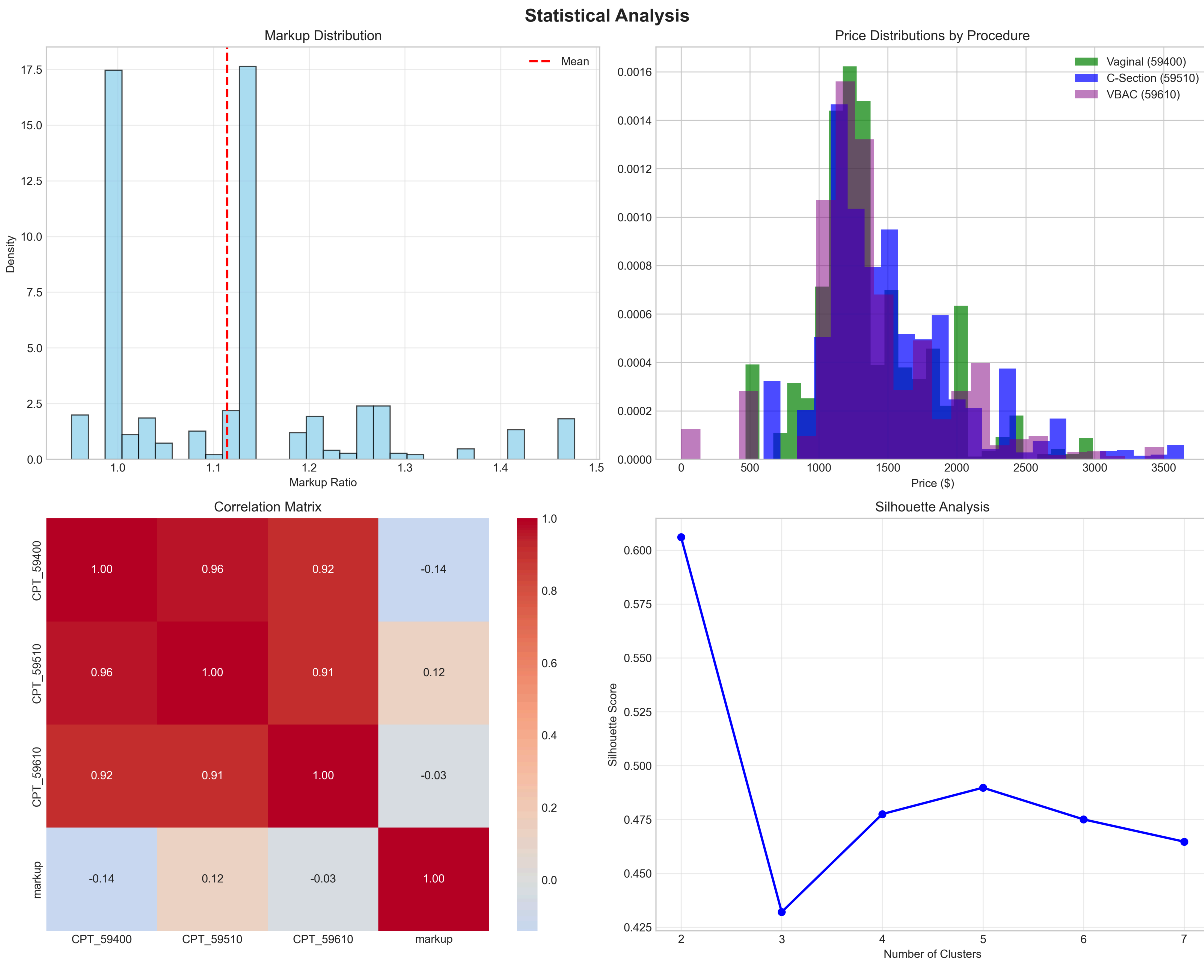
```
Medicaid OB/GYN Fee Schedule Analysis (1989–2012)
============================================================================
====
DATA QUALITY CHECK
Total obs: 15552
States: 54
NaN by CPT: CPT_59400    7758
CPT_59510    8145
CPT_59610    9613
dtype: int64
Time columns detected: ['Year']
Clean obs (all 3 CPTs): 5784 (37.2% of total)
After markup filter: 5166

==================================================
ENHANCEMENT 1: TIME TREND ANALYSIS
==================================================
Years available: [1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 199
8, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2
011, 2012]

Yearly Markup Trends:
       mean    std  count
Year
1989  1.139  0.000     12
1990  1.169  0.154     36
1991  1.169  0.154     36
1992  1.138  0.161     42
1993  1.105  0.140     60
1994  1.068  0.111     60
1995  1.057  0.105     72
1996  1.116  0.149    202
1997  1.115  0.147    246
1998  1.112  0.142    252
1999  1.115  0.141    252
2000  1.116  0.141    252
2001  1.122  0.135    269
2002  1.123  0.133    276
2003  1.121  0.132    280
2004  1.124  0.129    300
2005  1.115  0.132    312
2006  1.118  0.121    324
2007  1.115  0.112    324
2008  1.116  0.110    324
2009  1.119  0.101    324
2010  1.106  0.081    324
2011  1.103  0.082    313
2012  1.095  0.082    274

Annual C-section price trends:
        mean    std
Year
1989  2332.0    0.0
1990  1462.0  624.0
1991  1470.0  618.0
1992  1440.0  573.0
```

```
1993  1436.0  491.0
1994  1437.0  490.0
1995  1411.0  448.0
1996  1342.0  386.0
1997  1396.0  448.0
1998  1399.0  456.0
1999  1409.0  457.0
2000  1422.0  459.0
2001  1422.0  455.0
2002  1423.0  451.0
2003  1427.0  436.0
2004  1471.0  457.0
2005  1529.0  484.0
2006  1572.0  500.0
2007  1599.0  536.0
2008  1620.0  552.0
2009  1681.0  601.0
2010  1685.0  630.0
2011  1649.0  553.0
2012  1657.0  574.0

==================================================
State C-Section Markup Rankings (Top 15)
==================================================
            Avg_Markup Markup_Pct  Volatility  Observations
State
Minnesota        1.386     139.0%       0.189           203
Nebraska         1.354     135.0%       0.084           198
New Jersey       1.278     128.0%       0.000           204
Wyoming          1.250     125.0%       0.019           235
Colorado         1.200     120.0%       0.000           174
Wisconsin        1.191     119.0%       0.000           108
Montana          1.142     114.0%       0.058           204
Nevada           1.139     114.0%       0.000           288
Alaska           1.132     113.0%       0.005           173
Virginia         1.131     113.0%       0.006           204
Arizona          1.131     113.0%       0.004           196
Kansas           1.128     113.0%       0.000           204
Iowa             1.128     113.0%       0.118           276
Idaho            1.127     113.0%       0.011            84
Michigan         1.121     112.0%       0.006           137

Features shape: (5166, 3)
Any NaNs in features: False

K-MEANS CLUSTERING (Scaled + Complete Data)
Optimal clusters: 3

Cluster Centers (Unscaled):
         CPT_59400  CPT_59510  CPT_59610  markup
cluster
0           1023.0     1117.0     1033.0     1.0
1           2154.0     2433.0     2304.0     1.0
2           1429.0     1584.0     1459.0     1.0

==================================================
```

```
CLUSTER INTERPRETATION & LABELING
==================================================
Cluster ranking by C-section reimbursement (lowest to highest):
  Cluster 0: $1,117 - 1.11x markup
  Cluster 2: $1,584 - 1.11x markup
  Cluster 1: $2,433 - 1.13x markup

Cluster label distribution:
cluster_label
High-Cost States       821
Low-Cost States       2195
Medium-Cost States    2150
Name: count, dtype: int64

Sample state assignments:
      cluster_label                 State  count
0  High-Cost States                Alaska    173
1  High-Cost States               Arizona     77
2  High-Cost States  District of Columbia     28
3  High-Cost States               Montana     54
4  High-Cost States                Nevada    236
5  High-Cost States              Virginia     18
6  High-Cost States            Washington    138
7  High-Cost States               Wyoming     97
8   Low-Cost States               Alabama      7
9   Low-Cost States              Arkansas     93

==================================================
ENHANCEMENT 2: VALIDATED OUTLIER DETECTION
==================================================
Consensus outliers (2+ methods): 92 (1.8%)
Method agreement:
lof_outlier   -1     1   All
iso_outlier
-1            39   471   510
1            475  4181  4656
All          514  4652  5166

Top consensus outlier states: {'Wyoming': 61, 'Alaska': 31}

==================================================
VISUALIZATIONS
==================================================
```


Medicaid OB/GYN State Analysis
Top 15 States: Markup Ratio
Michigan
Idaho
Iowa
Kansas
Arizona
Virginia
Alaska
Nevada
Montana
Wisconsin
Colorado
Wyoming
New Jersey
Nebraska
Minnesota
Markup Ratio
Top 15 States: C-Section Price
Montana
Missouri
Minnesota
Michigan
Kansas
Iowa
Idaho
Georgia
District of Columbia
Colorado
California
Arkansas
Arizona
Alaska
Alabama
C-Section Price ($)
Top 15 States: Price Volatility
Volatility (Std Dev)
Top 15 States: Sample Size
Observations


Cluster Analysis & Outlier Validation
Cluster Centers by Procedure Type
Procedure
0
1
2
Average Price ($)
Vaginal
C-Section
Elbow Method
Optimal k=3
Inertia
Number of Clusters (k)
Outlier Detection Methods
Outliers Detected
iso_outlier
lof_outlier
z_outlier
consensus_outlier
Price Scatter by Cost Tier
Low-Cost States
High-Cost States
Medium-Cost States
C-Section ($)
Vaginal Delivery ($)




```
================================================================================
====
ANALYSIS SUMMARY WITH CLUSTER INTERPRETATION
================================================================================
====
Valid observations: 5166 across 27 states
Silhouette score: 0.432
Consensus outliers: 92 (1.8%)
Top markup states: ['Minnesota', 'Nebraska', 'New Jersey']
CLUSTER INTERPRETATION:
  High-Cost States: 821 states ($2,433 avg C-section)
  Low-Cost States: 2,195 states ($1,117 avg C-section)
  Medium-Cost States: 2,150 states ($1,584 avg C-section)
VISUALIZATIONS GENERATED (300 DPI):
   • medicaid_state_analysis.png
   • medicaid_cluster_outliers.png
   • medicaid_statistical.png

   • df_clean now contains 'cluster_label' column for further analysis
```

In [ ]: